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1. (4) +0.5

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<sol_start id=2>

2. (2) Outer surface of cornea

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3. (4) Virtual, erect & enlarged

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4. (A) Both A & B are true & R is the correct explanation of A.

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5. Assertion is true but reason is false.

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6. 2 common defects of vision :

(i) Myopia $\Rightarrow$ In this defect a person cannot see distant object clearly, also known as near sightedness.

Correct : He/She can wear concave lens.

- (ii) Hypermetropia : In this defect one can't see clearly the objects close to eye, also known as far sightedness.

Correction : One should wear convex lens.

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- 7.(a) A pencil when immersed in water appears bent & short because of the refraction of light. If an object is in denser medium observer is in rarer medium then the apparent height is less than the real height.

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- (b) Given

Object distance = $u = ?$

Magnification = $m = 20$

$v = -20 \text{ cm}$

$f = ?$

now using magnification formula

$$m = \frac{v}{u}$$

$$\Rightarrow 20 = \frac{-20}{u}$$

$$\Rightarrow u = -10 \text{ cm}$$

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8. Given

Object distance = $u = -10$ cm, focal length = 150 cm,

Image distance = $v = ?$

Now using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{15} = \frac{1}{v} + \frac{1}{(-10)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} + \frac{1}{10} = \frac{2+3}{30}$$

$$\Rightarrow v = 6 \text{ cm}$$

The image distance is negative that means image is virtual & erect & & f as it is convex mirror the image is diminished.

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9. Given

$u = -10$ cm, forms real image, $m = -3$

$v = ?$, $f = ?$

as it's forming real image, hence it's a convex lens & $m = -3$

now, using magnification formula.

$$m = \frac{v}{u}$$

$$-3 = \frac{v}{10}$$

$$\Rightarrow v = 30 \text{ cm}$$

Image is formed at 30 cm behind the sense.

now using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{30} + \frac{1}{10} = \frac{1+3}{10} = \frac{\cancel{4}}{\cancel{30}} \frac{2}{15}$$

$$f = \frac{15}{2} = 7.5 \text{ cm}$$

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10. Far point = x = 80 cm, f = p,

$\Rightarrow$ formula for focal length of lens for myopic lens.

$\Rightarrow f = -x \text{ cm}$

$f = -80 \text{ cm or } -0.8 \text{ m}$

Power of lens

$$\Rightarrow P = \frac{1}{f}$$

$$\Rightarrow P = \frac{1}{-0.8}$$

$\Rightarrow P = -1.25 \text{ D}$

The lens is concave lens

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11. (a) Convex lense

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(b) $\frac{v}{u}$ or magnification will be negative because it's a real & inverted image.

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(c) Given, $f = 20$ cm, $u = -21$ cm, $v = ?$

Using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{20} = \frac{1}{v} + \frac{1}{21}$$

$$\Rightarrow \frac{1}{20} - \frac{1}{21} = \frac{1}{v}$$

$$\Rightarrow \frac{1}{v} = \frac{21 - 20}{420}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{420}$$

$$\Rightarrow v = 420 \text{ cm or } 4.2 \text{ m}$$

The screen must be placed 4.2m or 420 cm away from the lens.

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12. (a) Laws of refraction of light.

- (i) The incident ray, the normal to the refracting surface at the point incidence and the refracted ray, all lie in the same plane .
- (ii) The ratio of sin of angle of incidence is equal to sin of angle of refraction for 2 given media

$$\Rightarrow \sin i = \sin r$$

$$\text{Absolute refractive index} = \frac{\text{Speed of light in vaccum}}{\text{Speed of light in medium}}$$

$$\Rightarrow n = \frac{c}{v}$$

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(b) Given with respect to

$$\text{Refractive index of water A/2} = \frac{4}{3}$$

$$\text{Refractive index of glass to A/2} = \frac{2}{3}$$

$$\text{Speed of light in glass} = 2 \times 10^8 \text{ ms}^{-1}$$

To find (i) speed of ligh in water & air

$\Rightarrow$ let the speed of light in water be S_1 , in air S_2 glass S_3

$$\Rightarrow \text{Refractive index of glass to air} = \frac{3}{2} = \frac{S_2}{S_3}$$

$$\Rightarrow \frac{3}{2} = \frac{S_2}{2 \times 10^8 \text{ ms}^{-1}}$$

$$\Rightarrow \frac{3 \times 10^8 \text{ ms}^{-1}}{4} = S_2$$

$$\Rightarrow \text{speed of light in air} = S_2 = 3 \times 10^8 \text{ ms}^{-1}$$

$$\text{Now, refractive index of water to air} = \frac{4}{3} = \frac{S_2}{S_1}$$

$$\Rightarrow \frac{4}{3} = \frac{3 \times 10^8 \text{ ms}^{-1}}{S_1}$$

$$\Rightarrow 4S_1 = 9 \times 10^8 \text{ ms}^{-1}$$

$$\Rightarrow S_1 = \frac{9}{4} \times 10^8 \text{ ms}^{-1} \text{ or } 2.25 \times 10^8 \text{ ms}^{-1}$$

$$\text{Now, speed of light in water} = 2.25 \times 10^8 \text{ ms}^{-1} = \frac{9}{4} \times 10^8 \text{ ms}^{-1}$$

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